Project Title:Secure File Upload, PDF Handling, and Malware Protection

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Category: Backend Security / File Security

Objective:
Research and design a secure system for handling uploaded PDFs, images, assignments, certificates, and institute documents safely. **Basic Project Approach**

1. Requirement Analysis
- Identify file types such as PDF, JPG, PNG, DOCX, and certificates.
- Define upload size limits for each category.
- Identify user roles and access permissions.

2. File Validation
- Validate file extension and MIME type.
- Restrict unsupported or executable files.
- Apply maximum file size validation.
- Rename files using secure unique names.

3. Malware Protection
- Scan uploaded files using antivirus tools.
- Detect malicious PDFs or infected files.- Reject suspicious or harmful uploads.

4. Secure Storage
- Store uploaded files in private storage.
- Avoid direct public access to files.
- Use signed URLs for secure file downloads.
- Maintain proper folder structure and permissions.

5. Risk Research
Study security risks such as:
- Executable file uploads
- Malicious PDF attacks
- Path traversal attacks
- Public bucket exposure
- Unauthorized direct file access

6. Secure Upload Flow
1. User uploads file
2. Backend validates file type and size
3. Virus scanning is performed
4. Secure filename is generated
5. File is stored in private storage
6. Authorized users access files through signed URLs

Expected Output
- Secure upload flow diagram
- File validation checklist
- Sample backend security rules- Risk and mitigation table

Conclusion
This project helps in understanding secure file upload handling and protection methods for backend systems. It improves application security by preventing malicious file uploads and unauthorized access.

Secure Upload Flow Diagram

User Upload
File Validation
Virus Scan
Secure File Naming
Private Storage
Signed URL Access